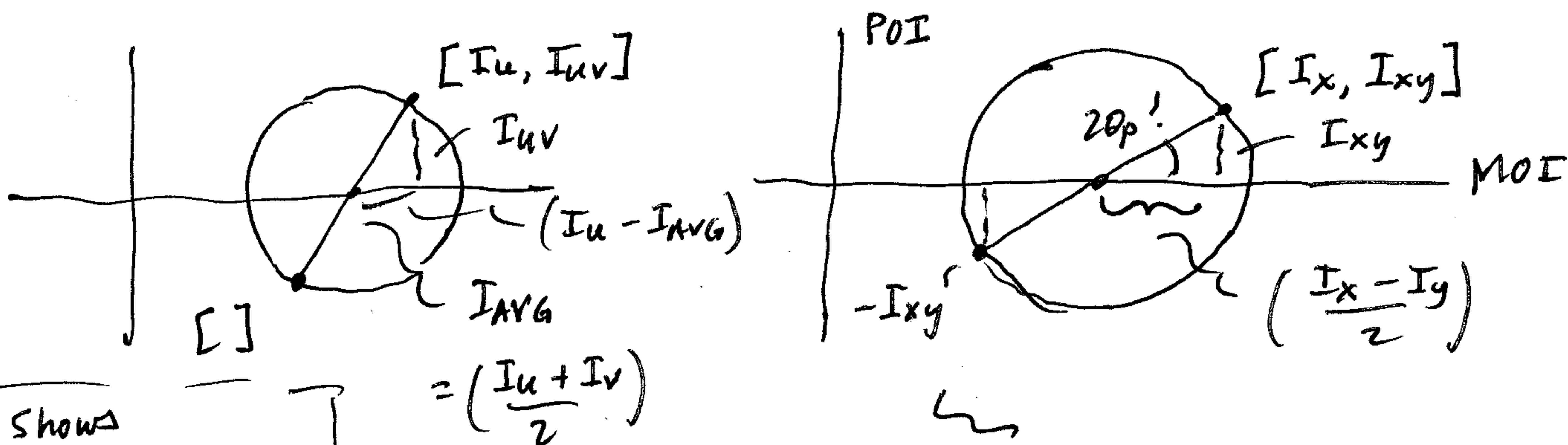


MC Derivation: End up with this:

$$(I_u - I_{AVG})^2 + I_{uv}^2 = \left(\frac{I_x - I_y}{2} \right)^2 + I_{xy}^2$$



MC shows us that I_{xy} can be +, -, or 0

$I_{AVG} = \left(\frac{I_u + I_v}{2} \right)$
MOI always +

Sim Δ's

$\left(\frac{I_x - I_y}{2} \right)$ same as $(I_x - I_{AVG})^2$

$$I_x - \left(\frac{I_x + I_y}{2} \right)$$

Know this!

$$= I_x - \frac{I_x}{2} - \frac{I_y}{2}$$

$$= \left(\frac{I_x - I_y}{2} \right) = I_x - I_{AVG}$$

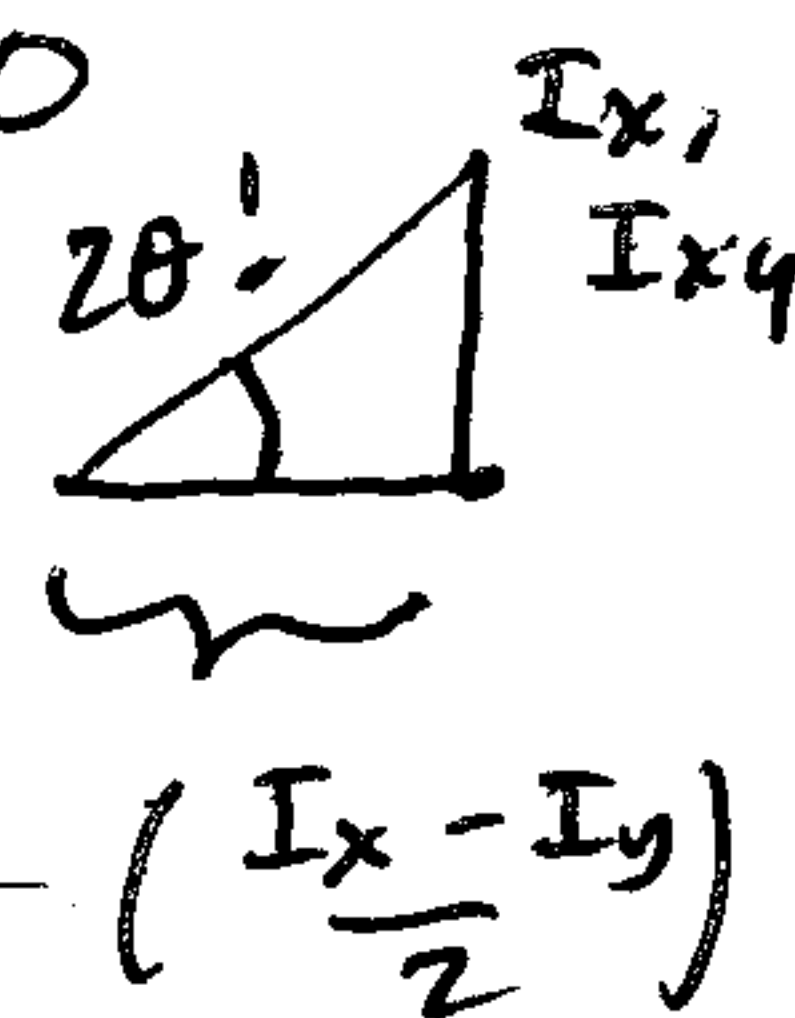
θ_p eqn?

$I_{uv} = 0$ at $I_{max} + I_{min}$

$$I_{uv} = () \sin 2\theta + I_{xy} \cos 2\theta = 0$$

$$\tan 2\theta_p = \frac{-I_{xy}}{\left(\frac{I_x - I_y}{2} \right)}$$

~ shows us that



* Nearest Principal MOI ...

How know if θ is to I_{max} or I_{min} ?

- Plug θ in I_u eqn
- Plot MC